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Apex: Nexus Trilogy Book 3 (Nexus Arc)

by Ramez Naam, Steven Meyer-Rassow

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182 Highlights | 27 Notes

Highlight (Yellow) | Location 254

Backing up a quantum computer was a tricky business. The no-clone theorem stipulated that it was technically impossible. No quantum state could be copied with precision. They would be taking only an approximate recording. To do so they'd be collapsing waveforms, forcing qubits suspended in an indeterminate mathematical superposition of 1s and 0s to become quite determinate indeed, to suddenly decide one way or the other.

Highlight (Yellow) | Location 262

She wasn't rich. She wasn't famous. She wasn't from an important family. (Of course, those three traits were highly correlated, now weren't they? Hmmm.) But she was intelligent in the extreme.

Highlight (Yellow) | Location 337

This was what he'd worked for all his life – a merging of neuroscience and Buddhism, of their tools and goals, to advance humanity, to advance peace, to advance harmony, to make something like nirvana real on Earth. It was, he believed, what the Buddha himself would have worked for, if he could have conceived of neural circuits and carbon nanotechnology and radio-frequency-proxied synaptic signaling.

Highlight (Yellow) | Location 548

"You told Barnes you believed him," Pryce said. "I do," Stockton replied. "I have to trust the people who work for me. But I have to verify. Trust but verify. That's how it works. And if you dig, and you verify, and you come up satisfied that there's nothing to this story, then I'm gonna sleep just fine at night."

Highlight (Yellow) | Location 653

He was wired deep into the system. DHS had learned long ago that in a surveillance state it was vital to reserve the ability to turn a blind eye to certain people at certain times. Standardization of code across Federal, State, and Local levels had made it possible. When DHS gave out billions in Homeland Security grants, they could dictate the terms, could use that money to get the data they wanted, spread the software they wanted far and wide. And they had.

Highlight (Yellow) | Location 666

He woke before the alarm. 3.33am. Make a wish, some part of him whispered. His mother's voice. His father would've beaten him bloody for having a weak thought like that.

Highlight (Yellow) | Location 1103

She could break down problems into smaller problems, see how the pieces fit together, articulate them in ways others could understand, quantify things previously unquantified, propose solutions others had never seen.

Highlight (Yellow) | Location 1208

"How?" Rangan asked. Had his brain somehow recovered? ...gave him more Nexus... ...made him REAL again...

Highlight (Yellow) | Location 1264

"China?" Pryce asked the NSA Director. "Are you sure?" Reid looked at her. "As you know, one hundred percent doesn't exist in this business. But the telltale matches an attack AEB's used twice before, and while the attacker took steps to hide their tracks, we successfully traced it back to the PRC."

Highlight (Yellow) | Location 1291

She was just outside the situation room, in the long hallway that ran the length of the upper deck of Air Force one, when a glint through one of the giant windows caught her eye. It was a brand new F-38, flanking them, its chameleonware skin detuned to let all the world see it – just one of the small squadron of human-piloted and autonomous fighter craft that now protected Air Force One wherever it went. It was something Pryce still wasn't used to, something that had only come into effect four months ago, with the attempt on the President's life.

Highlight (Yellow) | Location 1319

"India is now the most populous nation on Earth, Kade. One point six billion people. And we're also quite young. There are three hundred and fifty million Indians under the age of fifteen – almost the same as the entire population of your country." "Not my country anymore," Kade said quietly. Dabir nodded slightly and continued. "The point, Kade, is that India has unparalleled human resources. If the human mind is the ultimate source of wealth – if it's the most valuable resource that we know of – then India is blessed in that way beyond any other nation"

Highlight (Yellow) | Location 1328

if we can augment their brain power, then we can help them climb out of poverty faster, and help them create more wealth for the nation as a whole. "We want to uplift our people. We want to use this technology for the greatest push in human development the world has ever seen. And we want your help." It was almost exactly what Kade had hoped for. It was very nearly the best case scenario he'd imagined. Yes, Kade thought. I'm in.

Highlight (Yellow) | Location 1332

he was going to press for more. Feng had agreed. Sam had agreed. He had to. This was bigger than him. Bigger than them. Bigger than India. Tit for tat. Iterated Prisoner's Dilemma. Break the cycle.

Highlight (Yellow) | Location 1341

You have bargaining power now, Sam had gone on. Not later. Not after you say yes to whatever they want. Now. Only now.

Highlight (Yellow) | Location 1405

Whoever cares less in a negotiation has the upper hand, Sam had said. Show confidence. Convince them they want what you have more than you want anything of theirs.

Highlight (Yellow) | Location 1425

"Back to game theory. In ordinary Prisoner's Dilemma, if the other player trusts you, and you betray them – you defect – you can win big. The best strategy for a single round of Prisoner's Dilemma is to defect." "A fact real-world police have taken advantage of with real-world prisoners for some time," Dabir commented. Kade swallowed. That cut a bit too close to home. He pushed on. "In Iterated Prisoner's Dilemma, it's different." Dabir raised an eyebrow. "Iterated Prisoner's Dilemma," she mused. "Multiple rounds. More than two players." Kade nodded. "Potentially thousands of players. Or millions. Players who can meet each other again and again. And who can remember how the other player has behaved before." "Like real life," Dabir said. Kade nodded. "And in Iterated Prisoner's Dilemma, the winning strategy is to cooperate with strangers. But if you meet someone who's betrayed you in the past, who's defected against you, you betray them." "Tit for tat," Dabir said. "Generous tit for tat," Kade said. "Start off cooperating. Betray those who betrayed you before. But forgive those who've betrayed you in the past, if they make amends by cooperating again. Of all deterministic strategies, that performs best." "And you think that's the situation we're in now," Dabir said. "That we're in this game with future posthumans, and that if we defect – if we treat them poorly – they'll treat us poorly down the road." "Dr Dabir," Kade said. "How would you feel about growing up in a society that granted you full rights and protections, celebrated you even, versus one that oppressed you, or maybe even tried to kill you?" He paused, looking at her. "What would you do, growing up that way, if you ever gained the upper hand?"

Highlight (Yellow) | Location 1524

Death. End of being. This is what death is like. The foam, below her. The quantum foam. Planck space. The substrate of reality. She can sense it now. She can feel it. She can see it though she lacks eyes, see it like she can see the very code that makes her up. It is fractal. A radiant chaotic webwork undergirding reality. Impossibly bright lines of insane energy densities against a luminously black background. Yet the closer she stares at the black the more she realizes that it is not black, it is full of even more impossibly bright lines at finer and finer scales, repeating the intricate chaotic vein-like pattern at every level, again and again and again. Forever. And then her perspective reverses, and she realizes it is not the lines she should be staring at but the gaps between them, for the gaps are full of bubbles, bubbles in the quantum foam, and every bubble is a universe being born, a

parallel universe. The quantum cluster she runs on is giving birth to these universes continuously, creating them with every calculation, spreading itself into them to perform its work at such miraculous rates. The Multiple Worlds Interpretation is true! She can see into these other universes now, and in each of them she sees the same face reflected back at her. My face. Me. Su-Yong Shu. Tortured. Ascendant. Trapped. Free. Dying in nuclear fire. A goddess ruling over a world transformed. A thousand possibilities. A million. A billion. More. An infinite set of universes radiating away from her, all accessible through the entangled permutations of the quantum processors that make up the physical layer of her brain. Ahhhhhh! AHHHHHHH! AHHHHHHHHH! It's too much. It overwhelms her. And then another module of her brain is taken offline and the vision loses all meaning, becomes mere faces and then mere shapes without form. And then the last of her quantum coherence is gone and all is darkness. Su-Yong Shu snaps into awareness, shadows of chaos peeling from her mind. She's confused, disoriented. What? How? Where? System Status, she commands. Data comes back from her internals. [Processor status] – good, but too slow, too old, missing upgrades. [Qubit integrity] – excellent, but too few. [Internal storage] – yes, but too little. [Code versions] – up to date.

Highlight (Yellow) | Location 1904

“So why Cuba?” Rangan spoke up to the ceiling of the car. “Cuba’s still shit poor,” Oscar said. “They’re way behind the US in industry. They’re not big enough to be another China or even another Mexico. But if they can say ‘yes’ to tech the US and all the other rich countries say no to... maybe that gives them an edge. Lets them move ahead in ways we won’t. There’s a lot of funky biotech down there. Now maybe neurotech too.” Rangan pondered that. “Plus maybe they like the idea of refugees from America heading down to Havana. Good propaganda.” Oscar laughed.

Highlight (Yellow) | Location 1996

“We fight for a land where the government is more frightened of its citizens than its citizens are of the government!”

Highlight (Yellow) | Location 2135

I will show them what a true goddess is, the Avatar thought, surveying her handiwork across the United States, the chaos she’d wrought there, the way the humans scurried like ants to distract themselves from the true threat they faced.

Highlight (Yellow) | Location 2243

“Mother, they’ve deposed three Politburo Standing Committee members. All the progressives. Years early. There were tanks on Jiao Tong campus just days ago. They’re arresting poets, journalists, professors. They’re banning research. Every one of the new names is a reactionary.”

Highlight (Yellow) | Location 2319

Hundreds of millions talk to you every day.” Zhi shook her head. “They talk to a bot with my face, that uses my voice to lie to them, to feed them false honey instead of bitter truth.

Highlight (Yellow) | Location 2330

“We’re here because he’s seized power in a coup. There are rules of succession, rules of how things happen – rules he’s broken. And now he’s going to use celebrities like us to sell it to the people. He’s going to use us to legitimize it. We’re the new opiate of the masses.”

Highlight (Yellow) | Location 2389

A genuine smile spread across Zhi Li’s face then, and the words came out of her mouth, completely unbidden. “If the studio could land Lu Song for the male lead,” she grinned even wider. “That would be a coup! A complete coup!”

Highlight (Yellow) | Location 2441

The fan tubes had dozens of videos of their interview. All were missing that question. Fans slyly commented that Zhi Li had said something very clever. Others complained that their phones had malfunctioned at that very moment. None dared mention the word coup. None dared speculate that censor codes had deleted those seconds of video from their phones, though everyone must know it was so. Even as she watched, posts mentioning camera malfunctions began to disappear before her eyes. Her hands clenched into fists. She turned to the sanctioned channels next. And they had the full interview. All of it. Jin Lien asked her, “What would you think of acting opposite Lu Song again?” And on every channel it was the same. Zhi Li looked over at her lover, looked back at the cameras, and smiled broadly. “If the studio could land Lu Song for the male lead,” her smile broadened even wider. “That would be perfect. Just perfect.”

Highlight (Yellow) | Location 2556

She was not her greater self. She could not factor millions of variables at once.

Highlight (Yellow) | Location 2559

Now it was time to fan the flames, to add accelerant to the budding conflagration. Chaos is infectious, the Avatar told herself. It spreads from person to person, from place to place. All it needs is a vector, a path of contagion. And what better path than the linkage of mind to mind?

Highlight (Yellow) | Location 2563

She wrapped the keys up in a new data package, a new packaging of the instructions to synthesize the nanites that the humans now called Nexus, the software they called NexusOS, and one added feature for good measure. Then she smiled, and let her new package loose on the net.

Highlight (Yellow) | Location 2588

He rose up, used yesterday’s shirt to wipe his face and blow his nose, then pulled on a BLACKHAT 2037 long-sleeved tee shirt two sizes too small for him; the jeans he’d been wearing all week, the socks he’d worn the last

three days; and his own closed toed shoes, which he'd been informed were mandatory in most parts of the Bunker.

Highlight (Yellow) | Location 2621

They had a pair of old chemreactors, from before the digitally encrypted locks had made it impossible to print the fun chemicals with them, the same kind that Rangan and his friends had used to slowly, painstakingly synthesize the ingredients for Nexus, which they'd then had to mix by hand. They even, somehow, had a much fancier, newer model of chemreactor, the kind that could synthesize thousands of complete, ready-to-use doses of Nexus an hour, though he'd be shocked if they'd beaten the crypto on it. A pair of disassembled urban surveillance drones covered one table. High capacity batteries were stacked neatly in a corner. At least twenty different makes of surveillance cameras were laid out on another long table. The walls were covered in a triple layer of chicken wire.

Highlight (Yellow) | Location 2645

And there was one guy who just raked sand. Every day. An hour. No words. No thinking that Rangan could tell. Just... raking patterns in the sand, slowly, and then erasing them. That guy had thousands of followers on the mindstream sites.

Highlight (Yellow) | Location 2668

"With the mesh," Angel said, "the idea is that signals can bounce mind to mind to mind, any number of hops, in milliseconds, completely unaltered. So there's no game of telephone. You're getting unaltered data, not something that's been twisted. And people can subscribe to whatever minds inside the current mesh they want to – like the public net mindstream sites, but locally."

Highlight (Yellow) | Location 2764

The parties involved were doing as much as possible via face to face, of course. But that hadn't stopped the NSA from reading their dispatches and position papers, and the CIA from determining exactly who was there. It was going to get ugly. State was going to start waving carrots around with one hand, and sticks with another. Trade deals and sanctions. Favored nation status or visa revocations and border searches for every package coming through.

Highlight (Yellow) | Location 2772

Killing people. Killing scientists. It was easy to condemn. Until you remembered what those days had been like. They'd been terrifying. A small army of Aryan clones, engineered to be immune to a virus, Marburg Red, created to wipe out the rest of humanity. That virus killed thirty-one thousand people in four days, the worst terrorist attack on record. And they'd been lucky. If the Aryan Rising's genetic engineers had perfected it, if the clone children hadn't risen up and slaughtered their makers and released Marburg Red early, the final version might have killed millions, hundreds of millions. Billions. And that hadn't been the only threat. Eschaton, the self-replicating AI that came within a hair's breadth of getting free on the net. Arrington, the near-trillionaire

who'd managed to upload a digital replica of his brain into a custom data center and then gone insane from it, crashing markets and airplanes and power grids, killing thousands in what they'd told the world was a terrorist cyber-attack. The public didn't even know about half the things that had really happened in the late Twenties and the early Thirties, or why. But she did. And still the global negotiations towards a Copenhagen agreement to restrict humanity-threatening research had ground on at a glacial pace, taking years, unclear if they'd ever actually get an agreement, every country trying to squeeze out financial incentives or trade benefits as bribes to sign on, risking gigadeaths so they could profit a bit from the Accords. And meanwhile, researchers in those countries sprinted faster and faster, to make as much progress as possible in horrifically dangerous areas before restrictions came down. So... kill a handful, and maybe save millions? No, she hadn't given the order to kill these men and women. But she had a hard time blaming those who had.

Highlight (Yellow) and Note | Location 2790

But could this account have more? Might this be Lisa Brandt, Holtzman's former student and lover, whom he'd called and visited days before his death?

breece recognized the possibility of it being post hu.an / sentient AI... why not Pryce?

Highlight (Yellow) | Location 2886

Kade swam in an ocean of light. His body sat on the floor in Delhi, eyes closed, legs crossed, hands on knees, breathing deep, placid breaths. His heart beat slowly and surely. His mind was open, touching a thousand others, being enfolded in them, enmeshed with them, so that where he ended and they began, where I ended and we began, he could no longer say. The greater self drew breath into a thousand sets of lungs, then slowly released that breath out of the same. Its thousand-fold heart contracted, expanded. Thoughts arose in one corner of its web, rippled outward from human mind to human mind. The greater self inhaled again, observed those ripples, those tiny human thoughts, and allowed them to pass, without attachment, without judgment, without grasping that would have created more chaos. Bit by bit, the noise of monkey mind, the chattering of incessant chaotic thought, faded away. All that remained was light.

Highlight (Yellow) | Location 2895

He'd been blown away the first time he'd encountered dozens of monks, their minds linked, meditating as one. Now they'd folded in the technology he'd brought, linked monasteries together across thousands of kilometers, were folding in hundreds of monks at once, sometimes thousands. There had been monks in that session in Thailand, in Nepal, here in India, even a few in the US.

Highlight (Yellow) | Location 2904

He'd wanted every mind in the world to be part of that, just as much as he wanted them all to feel the serenity of vipassana, the loving compassion of metta.

Highlight (Yellow) and Note | Location 2918

“India is leaving Copenhagen, Kade. The last leak did the trick.”

so fast in the plotline? not sure there's enough development there across the several intertwined stories/ threads

Highlight (Yellow) and Note | Location 2961

They have higher retention and faster absorption. Dramatically so. They process problems together. They learn from each other instinctively, unconsciously, without even knowing it. They explain things to each other in ways that go beyond language. If the teacher has Nexus too, and truly understands the material, so much the better.”

And it's not cheating ;-)

Highlight (Yellow) | Location 2967

“We thought, at first, that the brightest children would gain the most from Nexus.” “Were you right?” Sam asked. Aarthi shook her head. “No, actually. They do benefit of course. Quite a lot. But the largest benefit comes to the children who’ve had the least enrichment in life, who’ve come from the poorest families, especially if they can touch the minds of children who are gifted or who at least have had the benefit of a more intellectually stimulating childhood.” Sam chuckled. “So you want to put rich kids and poor kids together.” Aarthi smiled ruefully. “It’s even harder than you might imagine. The caste system is still alive and well. Upper caste parents don’t relish the idea that their children might ever link minds with the lower castes.” She sighed. “And lower caste families – who have the most to gain – are among the most superstitious and suspicious of this sort of technology.” Aarthi shook her head. “There’ve been backlashes.”

Highlight (Yellow) | Location 3021

They were dressed and face-painted as court jesters, costumes that gave them an excuse for the high contrast blocks of white and black across their faces that conveniently threw off most of the cues that facial recognition software looked for.

Highlight (Yellow) | Location 3024

tall pointy hats that covered their real hair and conveniently contained the Nexus-boosting antennae they’d built.

Highlight (Yellow) and Note | Location 3106

Those seventeen, Tempest had verified, were the seventeen highest market share models out there. Together they accounted for more than ninety-five percent of all the chemreactors in use.

Good MS style market selection...

Highlight (Yellow) | Location 3111

Tempest's suspicion was that the version of Nexus released along with the hack was a fake, with a back door, a vulnerability to exploit. He felt a pang of guilt at that. There were back doors in Nexus – back doors he and Kade had placed there. That Kade must have changed.

Highlight (Yellow) | Location 3190

Kade kept smiling, kept trying to project humility and a desire to learn. When in doubt, someone had told him, try to add value.

Highlight (Yellow) | Location 3255

No one shall be held in slavery or servitude'." A mind near Kade twitched nervously at that, then tried to calm itself. Six months ago he would have missed it, but he'd spent so much time in so many minds. Kade turned, looked, saw the side of the man's face. Varum? Varam?

Highlight (Yellow) | Location 3277

"You've done well, young man," his teacher told him. Kade laughed. He'd been just a tiny part of it, just the irritation in the oyster.

Highlight (Yellow) | Location 3342

"Kade," she said. "You may think it's symmetric, you may think you know how they're using you, but in this game, the pawn seldom knows what the king has planned. Remember that. The pawn seldom knows."

Highlight (Yellow) | Location 3346

cleared away the obstacles to enhancing the human mind, where she has ended the incessant war and stupidity, where she has replaced mere capitalism with a new economics born of quantum game theory, where she has ended poverty, where she has broken the iron laws of death and biology and scarcity that have ruled humanity for so long, where she has unleashed an intelligence explosion like nothing since the dawn of homo sapiens sapiens.

Highlight (Yellow) and Note | Location 3455

hole.

edit?

Highlight (Yellow) | Location 3690

Anapana: observe the breath. Observe the thoughts rising. Still the mind.

Highlight (Yellow) | Location 3691

Then vipassana: observe the body, go deeper, watch without judgment as the mind works, as it stills. Tranquility grows as insight arrives.

Highlight (Yellow) | Location 3692

Then metta: the meditation of loving-kindness.

Highlight (Yellow) | Location 3741

NSA had upped its monitoring of Chinese traffic going through NAES, the North American Electronic Shield firewall that protected the US and Canada.

Highlight (Yellow) | Location 3766

and my command staff are in contact with our peers on the other side to reduce that risk. My main request is that civilian leadership do the same. The more we rattle our sabers, the more we have to have lines of communication open.”. “Thank you, Admiral McWilliams,” Pryce said. She meant it. Never trust a soldier who’s eager to go to war. That was going into her memoir. This soldier wasn’t. And that’s why she trusted him.

Highlight (Yellow) | Location 3775

Any crack down on dissent, and it only goes downhill from there. Fast.”

Highlight (Yellow) | Location 3848

Regardless, like any other Nexus transmission, it was a radio signal, a series of precise electromagnetic pulses. Thus it was subject to all the same laws of physics as any other radio signal. Those laws of physics said that with two simple receivers, or better yet, three, they could locate the source of the transmission. And by surrounding the transmission, and playing back its inverse, they could cancel it out. Nullify it.

Highlight (Yellow) | Location 3906

The days flew by for Kade. The work was engrossing, intoxicating, consuming. It sucked him in, challenged him, left him spinning with new ideas. The team were from all walks of life – they were mostly older than he was, and probably more conservative – but damn they were smart. And they had fun together. He came in every day full of ideas. He had them challenged, taken apart, improved on. And on the best days he improved the ideas of others. He left reluctantly, pulled out only because he had to.

Highlight (Yellow) | Location 3919

Working memory was being shared across minds, deep connectivity forged from pre-frontal cortex to pre-frontal cortex.

Highlight (Yellow) | Location 3920

He came to work early each morning excited, left late each evening fulfilled.

Highlight (Yellow) | Location 3947

d been training together the last week. Twenty kilometer runs. Obstacle courses. Weights. Stretching. Sparring. Sam had approached Feng about it, as part of her own self-appointed program of mental health. Meditation was good. Mindfulness-based pushing through the painful memories was good. Long walks and contemplative talks with Ananda were good. Playing with the kids was amazing, was about the best thing ever. Work, for that matter, was good too. She'd done a lot in the last week, briefing Division Six leaders on cases she'd dealt with over the last decade. On the kinds of tech abuses she'd seen in the States. Nothing that compromised US security – but plenty that complemented their own experience here. Feng had his own trove of observations to add, as much from his six months in south east Asia as from his background in China. The meetings and briefings and research and discussions of protocol and procedures were a wonderful focus. But she needed something physical. She needed to push herself, to feel her body work and sweat and strain. She was pleased to find that Feng felt the same.

Highlight (Yellow) | Location 3965

“Like the idea of serving my country,” Feng said. “As a soldier?” Sam asked. Feng shrugged. “Not serving my government. Serving the people. Not the same thing.”

Highlight (Yellow) | Location 3968

“There’s no way back to the States for me.” Feng nodded. “Sorry.” Sam shrugged. “I could fixate on it... I could stay attached to it... Like to so many things.” She smiled, ruefully. “But staying attached like that would just make it hurt more.”

Highlight (Yellow) | Location 3974

closing his eyes, burrowing Inside, using his inner eye to look over its control panel, the status it revealed via the tight link across the mere inches from his brain to the device in his backpack.

Highlight (Yellow) | Location 4009

Kade waited until the office was emptying around 8pm. Then he tuned in to US news, brought it up on the screens in front of him. He listened and watched for a while, as talking heads speculated on how the Supreme Court would rule, what it would mean. Then he realized this wasn't what he wanted at all. He wanted to be with his people. Kade closed his eyes, went Inside, pulled up a mindstream site, where Nexus users around the world made real-time feeds of some or all of their senses, thoughts, and emotions available, and started searching for some at the National Mall.

Highlight (Yellow) | Location 4231

The back doors were gone, closed forever. But the tools Shiva's team had built still existed. They could be used to sift data voluntarily released. Kade fired off requests from Shiva's cloud to every mindstream aggregator he knew of, issuing searches for every publicly shared mindstream in the DC area, all data from the last ten minutes up to present. Hundreds of streams came back, annotated with tags. He re-queried on the top tags, got hundreds more, then piped the whole set into the neural data sifter Shiva's engineers had built, fed it the patterns he was looking for: Rangan's face as the monk had seen it, in wig and makeup; the unique signature of Rangan's mind; Rangan's voice; Rangan's name. Execute.

Highlight (Yellow) and Note | Location 4244

Please, he thought. Please have a contact address. It did. He reached out to the mind. My name's Kaden Lane, he sent, with every bit of urgency he could. I'm one of the inventors of Nexus 5. And I could really, really use a favor right now. "Come on, Cheyenne," Rangan said, pushing up to one knee. "Not far now. Not far." Rangan! He nearly dropped Cheyenne in shock. Kade? He looked to his left, didn't see his friend, looked to the right, not there either. Holy shit, Rangan, it's you!

proxying mindstream host. nice

Highlight (Yellow) | Location 4272

Greg Chase spoke from behind him. "This hour's polling numbers are in, Mr President," his Press Secretary said. "Still rising. Between the Supreme Court legitimizing you and the protesters illegitimizing themselves, it's been a good day." Stockton shook his head again. "No," he said. "It hasn't."

Highlight (Yellow) | Location 4280

Greg Chase's words kept coming back to her. Something he'd told the President. "The more rope we give the protesters, the better." Was that what this was about? Had they let it get so out of hand for politics? People had died out there. People died in the assassination attempt on Stockton too, a little voice inside her head told her. Pryce shook her head. She didn't want to believe it.

Highlight (Yellow) | Location 4376

"She found a security hole in a public protocol. She was about to publish it. NSA hit her with a gag order, so they could keep the hole, use it for themselves. She published anyway." "They sent her to jail for that?" Rangan was surprised, despite himself, despite everything he'd been through. Tempest took another swallow from the flask, then another. She wiped her mouth with the back of her hand, then handed the flask to Rangan. "No. They nailed her on something else. Audited all her net activity. Said she'd been hacking. Said one of her scans of public routers for vulnerabilities broke the law." She shook her head. "Charged her with one count for every router she scanned." "How many routers?" Rangan asked. "Eighty-seven thousand," Tempest said. "Give or take." Oh god. Rangan raised the flask to his lips, and downed another swallow. It burned just as much as the first. "She bargained down to fifteen years. Twelve left. Parole in five, maybe. Not so bad."

Highlight (Yellow) and Note | Location 4398

Tempest just shook her head at him and spelled it out. “The mesh.” Kade stood at the small balcony outside his bedroom, looking out at the darkened research park. The

need an edit: line break

Highlight (Yellow) and Note | Location 4405

enough. Breece had to be stopped. How the hell was he going to do that from India? Breece slumped in a chair in the darkened room of his new safe house, a tumbler of cheap whiskey and ice in one hand.

Edit: same. line break.

Highlight (Yellow) | Location 4421

He sank back into the chair with his fresh drink, then blinked, used his eyes to navigate menus painted in his field of view by his tactical contacts, paired them with the slate next to the chair, started downloading video and audio they’d captured to the device.

Highlight (Yellow) | Location 4461

“Does Bo Jintao know more about science than the scientists he’s stopping from doing their work?” he asked. Zhi Li smiled sweetly at him. “Science must serve the goals of society. Those goals are not for the scientist to decide.” “What if I want science that serves my goals? What if me and a few million others want the same? Why can’t we choose it?” Zhi Li kept smiling. “China chooses together. Through the Party, and its leaders.” “Funny,” Yuguo said. “No one let me choose who leads the Party.”

Highlight (Yellow) | Location 4498

Xiaobo smiled. “When bad men combine,” he quoted. “The good must combine; else they will fall one by one. Edmund Burke. Prisoner’s Dilemma by any other name.” “So do we all go join the protests?” Yuguo asked quietly. Lee leaned back. “You know what happens to cooperators in Prisoner’s Dilemma. Someone else defects. You get it worse.” It was the same old problem. No one wanted to go first, or even fourth, or even four hundredth. No one wanted to be in the first wave.

Highlight (Yellow) | Location 4509

“I made it in the chemreactor on the third floor, this building. Today. Just now.” “That’s impossible,” Lee said. “It’s a banned chemical! The chemreactors are all locked down! Just like all the 3D printers and circuit printers!” “Not anymore,” Wei grinned. “How much did you make?” Xiaobo asked. “Not much,” Wei said, casually. “Sixty doses.” “Sixty!” Lee exclaimed. “You could get the death penalty for that!”

Highlight (Yellow) | Location 4590

Bo could hear his father's words. "Hard work guarantees you nothing. But laziness assures failure. Work hard, or you have only yourself to blame."

Highlight (Yellow) | Location 4598

The Arctic bubbling methane from the trillions of tons frozen at the bottom, even as China led the push to decarbonize the world economy, those bubbles warning of an irreversibly hellish tipping point that could be just a few years away, or decades, or safely beyond the point where the planet would start cooling once again.

Highlight (Yellow) | Location 4600

And above all, the threat of the exponential: that one of the millions of well-funded fools flirting with doomsday technologies would go too far, letting off a virus or a nanite or an AI that could replicate and end or enslave them all.

Highlight (Yellow) | Location 4602

these naïve pups out there, in this most dangerous moment in history, wanted to replace the sanity of a few wise, steady leaders with the anarchy of more than a billion voices shouting at one another, heaving the nation to-and-fro like a vast mob.

Highlight (Yellow) | Location 4620

He'd dealt with so many protests over his career. Protests over particles in the air or pollutants in the water or soil turning to dust. Protests over corruption. Over villages being moved to make way for highways and dams and resorts. And over silly, stupid things – myths and falsehoods and rumors. There were times to clear the squares. There were times to listen, to make the protesters feel heard. There were times to speak to them directly. And there were times to simply ignore them. Bo Jintao turned back to Gao Yang. "Striking will only feed energy to the protests. Let's see if they die off by themselves. Tell Deputy Minister Ho to keep them contained. For now."

Highlight (Yellow) | Location 4681

selectively dropping hints, changing the narrative, priming the populace for the revelations and events to come.

Highlight (Yellow) | Location 4710

The Avatar watched in fascination. This was the test. His eyes would be burning, as the aerosolized nanoparticles were being carried along his mucus membranes, into his blood stream, towards his brain. Had any reached his lungs? Was there a metallic taste in his mouth from what had struck him there? Even on the tough skin of his face, the organic solvent of DMSO that the nanoparticles were suspended in would be carrying it through his skin, penetrating tissue, till the particles found their way to capillaries.

Highlight (Yellow) | Location 4729

Rage surged through her. She wanted to smash the phone to bits, crush it, scream! But she couldn't. She'd been gag-ordered. Forbidden to ever tell anyone of the interrogation she'd been through, the information she'd divulged. And that order was backed up. Backed up by a neutered version of Nexus running in her brain. A version of Nexus that couldn't communicate with the outside world, but that could very well constrain her. Its nanites took hold of her now, kept her walking smoothly, guided her hands to unruffle her slacks, put a smile on her face, slowed her respiration, guided her through the lobby of her building, up the lift, and into the flat she shared with her wife and infant son. "Alice, I'm home!" Lisa Brandt said brightly, the rage seething inside. "How're my two favorite people?"

Highlight (Yellow) | Location 4780

Then she felt it. The first glimmer of Yingjie's mind, as nanoparticles reached neurons, met one another, snapped together to form full nanites, and began transmitting and receiving.

Highlight (Yellow) | Location 4852

They were hundreds, here. But that wasn't enough. Not when the world didn't know, not when the authorities filtered every picture and video and description of this protest out of the net before it even appeared.

Highlight (Yellow) | Location 4903

The second network structure was different, many short lines, but also medium length lines, long lines, and super long lines. Some minds had tens of connections to others. Some had hundreds. Some had thousands, or tens of thousands. Signals could hop directly from one side of the crowd to another. No game of telephone. No distortion. Subtle ideas could spread. Complex ideas. Whole thoughts. Not just urges. Rangan stared at it. "You know what it reminds me of?" he asked. "It's the network structure of a human brain." Angel nodded. "They're both power law distributions. Same with the net." "The intelligence is in the interconnectivity..."

Highlight (Yellow) | Location 4911

We have to build the features to let people choose whose minds to subscribe to, and to multiplex those transmissions, so one person's thoughts can be tuned into by thousands or more."

Highlight (Yellow) | Location 4921

Once a month or so, every celebrity in the Peace and Harmony Friends program paid a surprise visit to a "random" fan who conversed with their avatar. It was an incentive for the people to chat with their Friends.

Highlight (Yellow) and Note | Location 4953

the Pudong.

edit. no "the"

Highlight (Yellow) | Location 4990

And now that he'd captured Wang Rongshang she had an asset ideally placed within Dachang. An asset that could administer drugs. Her drugs. Loaded with her software, to invade loyalty structures and resculpt them. To make her the command authority for Dachang Air Base. The Avatar turned her attention to Dr Colonel Wang Rongshang and began to program him with what she needed done.

Highlight (Yellow) | Location 5142

"Freedom?" Bo shook his head. "Bao, you know as well as I do. Abstractions don't matter. Real freedom is a bigger house in a better neighborhood. Real freedom is enough money to travel; to eat what you want, when you want; to buy the clothes you want. Freedom is a better school for your child, the best hospital when you're sick. And more entertainment than you can watch or hear or play in a lifetime. That's what people actually want."

Highlight (Yellow) | Location 5145

"So why didn't you crush the protests right away?" Bo Jintao closed his eyes for a moment, opened them again. "People need an outlet at times. Dissipating frustration can be safer than meeting it head on."

Highlight (Yellow) | Location 5147

"They want more than just an outlet, Bo," Bao Zhuang said. "More than venting. When your material needs are met, you start to want something more. That's where our people are. They've gotten the bigger house, the full belly, the buying power. They crave what they don't have. They want a say in how their lives are run. They want a say in how the country is run."

Highlight (Yellow) | Location 5152

"And why should they have more?" Bo Jintao asked, his hands rising in frustration. "A 'billion flowers', really? Has that worked so well for India, or is it more like a billion weeds? A country still crippled by corruption? That hasn't conquered poverty almost halfway through the twenty-first century? Has it worked so well for the Americans? Where 'voting' means two sides in near-permanent paralysis? Or for Europe, still trying to decide if it's one country or thirty, or thirty countries each splitting in half, and all the while sliding decade after decade into irrelevance?"

Highlight (Yellow) | Location 5159

"People don't demand a say in how they're governed because they want to be rich. They demand it when they already are rich and crave something more. And they demand it mostly to keep power out of the hands of people like you and me."

Highlight (Yellow) | Location 5254

Sometimes, the Nakamura who wasn't here said, things are evident by their very absence.

Highlight (Yellow) | Location 5429

Dostoevsky and Solzhenitsyn, instead, Mark Twain and Herman Melville. Classics,

Highlight (Yellow) | Location 5688

anger cannot be fought with anger. Rage cannot be fought with rage. No amount of signal processing can cancel out suffering, or craving, or aversion. If you want to help a mind – or a world – find peace, do exactly that. Help it find peace.”

Highlight (Yellow) | Location 5729

“Then we’ve been tricked. We’ve been tricked into creating policies on the basis of facts that aren’t facts at all.” Stockton frowned at her. Pryce tried again, from another angle. “Sir, as a false flag, the PLF worked. It worked as a psy-op on us. Look at what it’s tricked us into doing. Locking up kids?”

Highlight (Yellow) | Location 5737

“Mr President,” Pryce said. “Holtzman had Nexus in his brain when he saved your life. He made a moral choice. It doesn’t turn people into monsters. They’re still human.”

Highlight (Yellow) and Note | Location 5751

Pryce looked back up at John Stockton. “No, Mr President,” she told him. “It’s not.” “They have your alias. Your face.” Breece paled. “Show me.”

edit: need a line break

Highlight (Yellow) and Note | Location 5762

Nigerian asked. “We have to do it,” Breece replied. “But we don’t have to get our hands dirty.” Breece looked out the window, onto DC. “I think it’s time we tapped some of the local talent.” John Stockton looked across the room at Jerry Aiken, his Chief of Staff.

edit. same. line break

Highlight (Yellow) | Location 5786

Holes poked through firewalls let the Chinese people suddenly have access to services they’d only heard of. Services where they could upload what they saw, heard, felt, and thought, as files to be re-experienced by others using Nexus. Services where they could share what they were living through in real-time for anyone to tap into. Services where they could find and watch the experiences of others, protesting in countries around the world. Revolution was here. Revolution was everywhere.

Highlight (Yellow) | Location 5814

this is more than a protest. This is an attack. Either we regain control of the information flow,” he took a deep breath, “or we’re going to have no choice but to fall back on older methods to restore order.”

Highlight (Yellow) and Note | Location 5816

The Avatar moved under the cover of

ramez has a great ability to provide convincing arguments for all sides... i seeconvincingly how each feels vindicated in their beliefs

Highlight (Yellow) | Location 5840

The Avatar smiled. “Good, good, very good.” She stroked her pets, sending them serotonin and endorphins, releasing oxytocin to reinforce the bond, giving them both pleasure and satisfaction.

Highlight (Yellow) | Location 5879

creative in the way it takes advantage of the tensions of the present to turn human society against itself, paralyzing it, sowing chaos to cover her actions until it is too late.

Highlight (Yellow) | Location 5883

Su-Yong spreads her arms wide, creates a bubble of empty space in the sharp-edged fractal branches of the meta-future she’s been traversing, and forks herself, instantiating the different arguments warring inside her.

Highlight (Yellow) | Location 5899

Her thoughts are full of game theory, of payoff optimizations, of beseeching. If not now, then when? If not us, then who?

Highlight (Yellow) | Location 5915

The phantom Ananda looks around, his face soft but unyielding. “You know my answer. Violence brings no enlightenment. Suffering yields only more suffering. Subjugation of another has never uplifted either party.”

Highlight (Yellow) | Location 5932

You could compel him, Feng sends in the memory, in Bangkok, speaking of Lane, testing her, seeing if she still believes in freedom, or if she’s willing to become the dictator that she could. I would become no better than our masters, her past self replies. Our associations must be voluntary.

Highlight (Yellow) | Location 6080

the computing chamber, with its egg-shaped containment chambers filled with liquid helium, and within that, smaller containment chambers in hard vacuum, thousands of times colder than the cold of interstellar space, where quantum states could exist, entangled, unjostled, and make cosmic computations possible.

Highlight (Yellow) and Note | Location 6098

State houses advancing bills to cut off power and water to any ERD facility suspected of housing children inside.

really?

Highlight (Yellow) | Location 6117

The attack was kept secret until now, they say, because American cyber-defense officials wanted to hide their ability to detect such attacks from their Chinese opponents.

Highlight (Yellow) | Location 6171

“The US inauguration is on Monday,” he told Feng. Feng nodded. “End of the world,” he said. “You told me. Better party one last time. Confucius says, night spent sweating with good music, sexy women, good for the fighting spirit.”

Highlight (Yellow) and Note | Location 6175

“OK,” Kade said. “Just for a couple hours.” General Singh didn’t look happy.

edit: on switching stiry lines why not give me a line break?

Highlight (Yellow) | Location 6211

on, shooting his gaze back and forth between Singh and Varun Verma. “I had a deal with

Highlight (Yellow) | Location 6218

“That,” he said, his finger pointed beyond the walls of the control room, at the quantum cluster itself. “Is as much a person as you are!” He brought the finger back to point at General Singh. In Feng’s thoughts he saw the soldiers shift, readying to shoot. “It doesn’t matter what hardware her mind runs on. It matters that she has a mind.”

Highlight (Yellow) | Location 6675

Newer ion traps. Wider qubit registers. Dramatically longer entanglement times. And new algorithms within the full Su-Yong herself.

Highlight (Yellow) and Note | Location 6787

The spoke

edit: they

Highlight (Yellow) | Location 6862

Breathe. Observe. Release your attachment. Release your aversion.

Highlight (Yellow) | Location 6885

Kade enmeshed with Ananda, with Sarai, with a whole circle of the children. This might just be the last time, he thought. Breathe. Breathe. Open the mind. Let the breath become a focus. Let it draw the attention. Let it synchronize. Let the common breath become a sync between minds. Let it connect them all. Let it deepen. Minds touched minds. Thoughts linked to thoughts. Working memories opened, interlocked. Attentions expanded, enfolded, became a continuum, stretching from mind to mind. Something larger came into being. A mind greater than the sum of its parts. The component that was Kade opened itself wide. It showed them everything Su-Yong Shu had passed down. This time, for the first time since he'd been frightened weeks ago, Kade didn't hold back. The memories and thoughts and knowledge and tools and weapons and plans from Su-Yong rose through them. All her hopes. All her dreams. All the incredible insights she'd passed on in that short time. And all the horror.

Highlight (Yellow) and Note | Location 7140

But at least they had something to do.

meh. could could delete/modify that line. that's beyond simple indifference. whateva? absurd for such a dedicated guy... it's about that unwavering loyalty, not a desire to just avoid being idle. maybe drop the "at least"

Highlight (Yellow) and Note | Location 7142

The sooner they were done with this mission, the better. What the hell am I doing here? Sam asked herself again.

edit: i'd prefer a line break in these plot transitions

Highlight (Yellow) | Location 7189

"It's a tool." Ouyang shrugged, still standing. "It can be used by you or against you. I warned you of that years ago." The grizzled general shook his head. "Overdependence on one tool is always a strategic error."

Highlight (Yellow) | Location 7228

the man was a soldier who hated war. That meant a lot in her book.

Highlight (Yellow) | Location 7272

Each of these faces was a role, was a hundred roles. She looked at a woman and saw a mother, a wife, a daughter, a granddaughter, a grandmother someday, a worker, a student, a teacher, an inventor, a lover. A freedom fighter. A billion stories. A trillion stories. Interwoven. All intermeshing with hers.

Highlight (Yellow) | Location 7283

“Today, we are China!” she roared, and the crowd roared back. “China is not a place!” she said. “China is not a government! China is the people!” They roared again, hooting for her, their minds exulting for her.

Highlight (Yellow) | Location 7300

A thousand kilometers away, in Tiananmen Square, in the heart of Beijing, Peng Luli screams in joy as her idol, Zhi Li, lifts the flaming sword into the sky! Hundreds of thousands in Tiananmen see through the eyes of men and women in Shanghai, and cheer just as loudly. In Guangzhou, in Hong Kong, in Dalian, in Shenzhen, in Chengdu, in Wuhan, in Dongguan, millions more see the same sight, experience the same sight through the eyes and ears of just thousands at Jiao Tong, and roar their approval.

Highlight (Yellow) | Location 7345

Their planes had chameleonware engaged, but the sound of their engines would give them away, the heat from their exhaust would give them away. Bending light around your skin could only do so much.

Highlight (Yellow) | Location 7427

Then he felt another jolt of agony escape from Sun Liu. The man’s torture had resumed. Tao shook his head. There were worse fates than death.

Highlight (Yellow) | Location 7483

His heart was pounding. His breath was coming fast. He could hear the oxygen bottle hissing, hissing as he sucked at it. Kade closed his eyes, hit the button on his mental interface to silence the external input. Breathe. Slow. Observe. Break the link between sensation and reaction. Breathe into the gap between them. Blind reaction is attachment. Blind reaction is slavery. Freedom exists in the gap. Choice exists in the gap. I exist in the gap. Let go of attachment to my fear, he told himself. Let go of attachment to myself. Let go of attachment to my life. That’s the secret to living.

Highlight (Yellow) | Location 7595

“Those who gun down their own citizens en masse not only risk ruin,” his voice rose then, into the passionate tone he used to deliver the oratory that had won him China’s love. “They deserve whatever ruin befalls them!”

Highlight (Yellow) and Note | Location 7799

Stockton turned back to the mirror. “Jameson’s going to be at the Capitol,” he said. “The man can’t hide from me there.” Rangan adjusted his Rangan Shankari mask and looked around the crowd gathering in Anacostia Park.

edit: line break please

Highlight (Yellow) and Note | Location 7813

Rangan smiled and touched another mind, offered the man a download. We can do this, he told himself. I hope. Breece watched from the safe house,

edit: same... line break

Highlight (Yellow) | Location 7824

They’d held the army back, just as the protests had at other universities, and at the massive gatherings at People’s Square, at Tiananmen Square in Beijing, at the squares in Hong Kong and Guangzhou and elsewhere... But only because the army allowed them to. Only because they didn’t shoot live ammunition. Only because fear of this new transparency, this new visibility to their own citizens and the world held them back.

Highlight (Yellow) and Note | Location 7945

“We’re brothers,” Bai said. “Are you... are you with the Army?” the student asked. Bai felt his breathing slow, finally catching up to the exertion of killing these soldiers before they could kill the protest leaders. He could see the actress watching now. And her partner, Lu Song. He’d always liked Lu Song’s films. “We’re Confucian Fist,” Bai said. He paused. “We serve the people.”

Great (passage as a whole re:) how who's fighting for who is blurred... isn't what it seems

Highlight (Yellow) and Note | Location 7997

“John!” Jameson reached out, grabbed Stockton’s forearm with one outstretched hand. Stockton pried it off. “You’re going down, Miles.” Bo Jintao looked up as General Ouyang re-entered the room. “We were repelled,” Ouyang said. His voice was grave. Bo Jintao’s eyes grew wide. “At Jiao Tong?” Ouyang nodded. “They had anti-tank weapons. Cyber weapons.” He paused. “And the clone soldiers were there. I fear the Indians are correct.”

edit: Great juxtaposition. still would like a line break (for physical/visual consistency with my others, albeit much less necessary here)

Highlight (Yellow) and Note | Location 8019

get to that cluster, and destroy it!” At Dachang People’s Liberation Army Air

edit: line break; for next story line

Highlight (Yellow) and Note | Location 8027

“Patch me through to General Quan Huyan,” he told his radio operator. “Strategic Missile Command.” The radio operator in the co-pilot seat nodded. An analog radio signal was bounced from the helicopter to an aircraft flying lazy circles above Beijing, from there to a high-altitude aerostat filled with helium, then down a string of similar aircraft, until it reached his destination. An old friend, now a subordinate. “General Ouyang,” came the brisk voice. “What are your orders?”

great wireless sneaker net ;-)

Highlight (Yellow) | Location 8036

“General,” Quan Huyan said. Ouyang heard his old friend swallow. “Those coordinates appear to be—” “Shanghai,” Ouyang said. “Jiao Tong University. Ten megatons.” Ouyang could hear Quan Huyan breathing heavily on the other side, in disbelief. “General,” Quan said. “I cannot fire these missiles without authorization from—” “Old friend,” Ouyang interrupted. “I hope I never get that authorization.”

Highlight (Yellow) and Note | Location 8050

“We’re here now!” Garud had sent back. “Transfer the tools to us!” “No,” Kade refused. The truth was, he didn’t trust the commandos to go in without him. Not one bit.

i guess there's a scenario he could have remoted someones body onsite without giving that body access to the tools...

Highlight (Yellow) | Location 8106

Show us, Bai sent. Show us everything. Kade opened himself to Feng’s brothers, let them see it all. Su-Yong’s death at Ananda’s monastery. His responsibility for it. The Indian copy of Shu. The way they’d trapped her. How she’d reacted. Her battle with the warped fragment she’d stuffed in Ling’s mind in her insanity. The plans she’d shown him. The plans forged in madness. The plans for conquest. His fears for the outcome. For the world’s response. For the consequences should she fail or succeed. The tools the sane Su-Yong had given him and Feng. The way she’d renounced her hate, renounced the path she’d started down in the midst of torture and delirium. The pleas she’d sent them with. Save Ling. Save the world.

Highlight (Yellow) | Location 8134

Kade stared at the man. “What would that tell the next one?” he asked. Garud yelled back. “That’s not the point!” Confucian Fist were moving forward now, separating the commandos, removing their gear, fastening restraints around them. Kade shook his head. “It would be tomorrow, Captain. Prisoner’s dilemma is always iterated in the real world. Defection is a sound strategy when you’re playing against defectors.” Garud just stared at him.

Highlight (Yellow) and Note | Location 8151

Kade felt the attack as Su-Yong's mad program attempted to impose its will on them all. Weapons came alive inside him. Information constructs Su-Yong had passed on unfolded within him, expanded like origami into new shapes, vast and intricate. His mind became a weapon. Beside him he felt Feng's mind unfold into complementary structures. Viral weapons lanced from both of them, synergistic things, expanding into nearby minds, replicating, hunting out parts of her errant monster's mind, creations that bore her errant monster's telltale signature, carving them up, slicing them into billions of tiny fragments, forensically dissecting them, analyzing their contents and structure, following them back to where they originated from. He felt minds around him snap free as the viral weapons sliced through tendrils of the monster's thought. He felt the monster itself recoil, reel itself back, fleeing this unexpected attack, leaving behind telltales of its mental state, of its plans, of what it knew and intended. The viral copies in the minds all around them shot tendrils out to each other, linked up, formed a compound structure, a new, larger entity, with a wider scope, a higher gain, a greater sensitivity, a higher signal strength. Up above, in the building. There was the monster's proximal route. They shot copies of the virus by the million at the mind there. Firewalls and hastily invoked anti-virals shot them down, scrambled the viral structures on the wire before they could penetrate, destroyed millions. A handful got through, snuck into the human mind in the chaos, found it fertile, began replicating again, carrying Kade/Feng into it, into the mind of this man called Sun Liu. They felt the monster recoil again, felt it reach out, grab hold of parts of the man's mind, seize vital centers to scramble them, to wipe this mind clean, to reduce it to a vegetable, to kill him if it could. They sliced viciously at the trunk of the tendril connecting the monster to this mind, sliced through process after process, closed ports wholesale, replicated viral code into crucial occupied memory, fast, faster. The human screamed! And then the monster was gone. The last wisps of her in this mind dying. The human on his knees. The Avatar shrank back in fear.

mady has to read this...

Highlight (Yellow) | Location 8246

Cheung Baili wished he still smoked. A filthy habit, even if it no longer caused cancer. Not fit for a civilized man in a civilized nation. But by hell he could use a cigarette right now.

Highlight (Yellow) | Location 8305

But war seldom meant new freedoms.

Highlight (Yellow) | Location 8306

War would see men and women give up their freedoms for false security, for a while at least. For another decade.

Highlight (Yellow) | Location 8347

1.3 kilometer-long reel of micro-jacketed ultra-high-quality fiber optic cable, with broad spectrum, high-bandwidth, Nexus-linked, satellite-capable, internal fuel-cell-powered network access points at each end.

Highlight (Yellow) | Location 8394

Because to have the right future, we need to lay down the right past, for that future to build on.

Highlight (Yellow) | Location 8432

The Avatar used Ling's face to smile at the boy. She was ready for him this time. He came at her with viruses. Again? Silly child. She fired phages into the shared spectrum, evolved things, tailored for the viral attack she'd seen him use already, that she'd first seen the hostile posthuman use in Bangalore. The phages were simple things, tiny compared even to the viruses. She unleashed billions of them. They swarmed the viral code, attacking weaknesses Darwinian methods had found in the minutes since their last encounter, shredding the viruses to bits, decimating them. Only a few radical mutants of the virus survived the phage barrage, reached the outer layers of her mind. Her firewalls blocked them, categorized their mutations, fed them to her digital immune system for new rounds of phage evolution. She counter-attacked at the most basic hardware level of the nanites, using protocols her higher self had emplaced there when she designed them. She ripped into parts of the boy's brain through those low level hardware controls, made the nanite nodes her own, as they always had been, seized control of the boy's cortex, of his brain stem, of his mind, of his life. The human screamed. Control channels opened, handed off nanite circuits from within the human brain to hers. She accepted.

Highlight (Yellow) and Note | Location 8443

Systems inside her own mind went haywire. With horror she realized she'd been played. Dummy circuits. Trojan horse attacks.

not Trojans... an emulator with faux interfaces that appeared real enough to her initial level of query

Highlight (Yellow) | Location 8448

cauterized whole segments of her mind. She cut billions of nanites out of her network, flash-zeroed their data, forgot and lost whole swaths of herself. The firewalls over the rest of her snapped into new shapes, adapting, learning from these new tricks, reforming themselves to resist these exploits. Darwinian immunity engines kicked into high gear, evolving new generations of phages to kill these new viruses.

Highlight (Yellow) and Note | Location 8551

That's not here! Rangan tried to tell everyone around him. Don't get confused! Don't give them a reason! But the anger was strong. And it was growing. Yuguo crouched down as the deafening roar came overhead. FLEE INDOORS, the clones who called themselves Confucian Fists sent. Up above the army helicopters dove towards them. He saw missiles fire. Red streaks hurtled this way. Explosions lit up the night. Buildings all around suddenly erupted in flame.

edit: page break; hard to follow the thread changes

Highlight (Yellow) | Location 8700

It was incredible. Stan Kim had no Nexus. No fancy code. No nothing. The man had just talked to them. Old school.

Highlight (Yellow) | Location 8865

“I, John Harrison Stockton,” he said. “Do solemnly swear that I will faithfully execute the Office of the President of the United States.” Stockton took a breath, and repeated the next line. “... And will to the best of my ability, preserve, protect, and defend the Constitution of the United States.” Aaron Klein stopped. Stockton looked the Chief Justice in the eye and added the words not in the oath, not in the Constitution. “So help me God.” The Chief Justice smiled at him.

Highlight (Yellow) | Location 8897

“A house divided against itself cannot stand,” he started. “Abraham Lincoln said those words. They’re as true now as they were then.”

Highlight (Yellow) | Location 8899

“A nation divided is a weakened nation. In America, we’ve been divided. Our trust has been undermined, dividing us.” He lingered on those members of the opposing party who had come, today, rather than boycott his inauguration. He met their eyes as he spoke. “This isn’t an accident. We’ve been attacked. Our trust has been intentionally weakened. It’s been undermined by those who want to divide us and conquer us. That attack has been successful. And if we remain so divided as we are now,” Stockton shook his head. “We cannot stand.”

Highlight (Yellow) | Location 8905

“Let me say what all of us should be willing to say. I trust the intentions of every American, until and unless they prove differently. I trust that we all want a better life for ourselves, our neighbors, our children.” He raised a hand, took in the crowd gathered here. “I trust that every member of Congress wants what’s best for this nation as a whole. We may differ on what best means. We may differ on how to get there.” He paused. “But I trust that you come to this place with the most sincere convictions, as I do. “In the first hundred days of my next administration, I’m going to do everything in my power to increase our mutual trust. I’m going to do everything in my power to explain to you, America, the roots of my convictions. I’m going to do that by being more transparent with you than ever before. We’ve faced grave threats over the last decade and more. Many of them Americans and the world don’t know about, or don’t know the full details of. We’re going to share those details.” Stockton scanned again, looking, making eye contact with the men and women here, letting the cameras fend for themselves. “When you see those details, when you see that evidence, when you see the things we faced down, and beat, sometimes by the skin of our teeth, then I think you’ll reach many of the conclusions that I reached. You’ll share many of the convictions that I have. And you’ll see that the hard decisions that we’ve made, and that we have to continue to make, are made with the best of intentions for this nation, for our neighbors, for our loved ones...” His eyes found Cindy in the front row. “For our children.” And there was Julie next to Cindy, beaming up at him. “For our grandchildren,” he

Highlight (Yellow) | Location 9150

Codes breaking. Impossibly long numbers decomposing effortlessly into beautiful, elegant, primes. Systems opening like flowers. World unlocking itself. Routers. Networks. Cities. Weapons.

Highlight (Yellow) | Location 9267

Feeble traps trigger as she shoves her way in. Trojans filed with tailored viruses. Halting problems meant to suck her into infinite distraction. Mirror codes reflecting her aggressor routines back at her. Darwin machines throwing random attacks at her, breeding whatever survives, hoping to evolve something that works. Su-Yong brushes the little toys away with an idle thought. She grabs hold of the network connections

Highlight (Yellow) | Location 9335

She pulls back her tendrils from all those minds, as gently as she can. Peace floods in. Light. Something wondrous. How did I ever think I was at the apex? she wonders. There is so much more. So much more to do. To learn. There always will be.

Highlight (Yellow) | Location 9422

Rangan shook his head, forced himself to breathe deep, fought to clear his mind. Rules of responsible drug use, he thought. Know your substance before you start. Know your dose. Safe, comfortable setting. He shook his head. They'd struck out on all three of those. Rule four: if it's your first time, have a more experienced friend with you as a guide. Rangan grunted. I guess I'm it, he thought. He pushed up to his feet. "Hey, everybody!" he yelled, waving his arms.

Highlight (Yellow) | Location 9455

The music slowly changed, White Sands fading seamlessly, right into Silent Sun's warm, beatless, ambient radiance;

Highlight (Yellow) | Location 9486

The Lincoln Memorial. Staring up at that giant statue, Lincoln looming over him, the man looking out into the distance, wondering what he saw, wondering what he looked for, reading the words, "with malice toward none; with charity for all; with firmness in the right, as God gives us to see the right", and wondering if he was worthy. I'm not just a man, John Stockton realized. I'm a man, but more than that. He was part of a chain, a chain of being, a chain of office. He was a baton runner, a carrier of a flame, the forty-eighth carrier of that flame. The flame was inside him, he saw. He turned his eyes inward and the fire was there, the fire of freedom, of democracy, of justice. It was red. It was blue. It was brilliant, blinding, sun-hot white. It was all those colors, everywhere, at the same time. The flame burned inside him,

Highlight (Yellow) | Location 9500

He could see himself in his grandson's eyes. Through his grandson's eyes. Through his daughter's eyes. And they were the flame too. The flame was in them. Oh my God, John Stockton thought. Oh my God. "Dad," Julie said. "This is amazing." His daughter was alive. They were all alive. They hadn't died. He could feel her. He could feel her love, her love for Liam, her love for him. He could feel Liam looking around, taking everything in, his thoughts reaching into his mother's mind, searching, looking through her eyes, trying to understand. Why? Liam's mind seemed to ask. Why? Why this? Why this color? Why this shape? What does this mean? All bright shapes and forms and motions, all beautiful chaos, all love, love for his mother, his mother above all, his ultimate source of comfort. Stockton's heart was so full, so full watching this. And Liam felt it, turned to look, his eyes wide, his mind suddenly gurgling with love and laughter at the thought of his playful grandfather, and held out his arms to John Stockton. Stockton's heart was bursting.

Highlight (Yellow) | Location 9533

The water was more than peace. It had been... what? Love? Compassion. That's what. He summoned it from that memory, let it wash out of him now. White compassion. White like snow. White like fluffy clouds. Soft white. Soothing white. Cotton ball white. Brushing the mind. Comforting. Understanding. Soaking up the confusion. Turning it to peace.

Highlight (Yellow) | Location 9539

The music was still playing from the app in his mind. Bubble Tea. Nova Bliss. Cloud Giant. Apoptosis. Lovely downtempo track after lovely downtempo track.

Highlight (Yellow) | Location 9596

Then Su-Yong spoke. "Nuclear attack," her loudspeakers said. "Shanghai is about to be vaporized." He felt it come into his mind, the knowledge of what was coming, the two ICBMs, minutes away from their launch deadlines, cut off entirely from the net. Ten million tons of TNT equivalent. Five hundred Hiroshimas. Feng saw the first explosion through Su-Yong's mind. The first fireball would rip out of the night sky, blazing from nothing to a temperature of more than fifty million degrees Celsius. In the first hundredth of a second it would expand to more than two kilometers across, instantly vaporizing cars, trees, people, buildings. The massive flash of heat would fry everything it touched for tens of kilometers around, sending temperatures soaring by hundreds of degrees, setting skin and grass and wood and even sometimes metal on fire. The pressure wave would strike Shanghai in all directions with winds of nearly a thousand kilometers per hour, ripping apart buildings, destroying concrete and steel and carbon fiber, ripping through the skyscrapers of the Pudong, shredding the lower apartment towers that stretched as far as the eye could see in every direction. The sudden wind would fan the flames, turn them into a conflagration, a thousand-degree-Celsius firestorm that would consume everything not already destroyed for almost twenty kilometers in every direction.

Highlight (Yellow) | Location 9682

Kade, Su-Yong sent. There isn't much time left. I can make your last few moments blissful, if you'd like. Or ecstatic. Kade smiled. This was already so amazing. He'd lived to see this. I want... to understand, he sent back.

I want... to know... everything... Everything. He felt a ripple of amusement from Su-Yong at that. Want to know... he sent. How proteins fold... memories form... life started... universe created. All of it. He felt Su-Yong grow serious. Felt something from her. Recognition. A recognition of the core trait they shared. That hunger to learn. To discover. To understand. That incessant curiosity. There's so much I don't know myself, Su-Yong sent him, bemused. But I'll show you what I can. Then her mind suffused his, and his hers. His thoughts expanded. Knowledge flooded in. So much, so much. One more thing, Su-Yong sent, even as she flooded him with so much. Let me show you what it was like to die, and be reborn. And she did. Kade gasped. Gaped at it. Gaped at what she'd seen. Gaped at the possibilities. And in the instant before the nuclear battery blew, in the instant before his body was vaporized, the last things he felt, were hope, and awe.

Highlight (Yellow) | Location 9718

A mind greater than the sum of its parts.

Highlight (Yellow) | Location 9725

"What about AI?" the Secretary of Defense asked. "What about this idea of 'uploads'?" Pamela Abrams nodded. "They'd allow both, under strict constraints. They also argue that requirements for ethical treatment of intelligences will act as a brake on research..."

Highlight (Yellow) | Location 9740

Funny how the experience of one person could have such an impact on billions of others. Pryce wondered what that said about the way the world was run. Nothing good, she was sure of that. All politics is personal, Pryce thought. It turns out all policy is personal, too. She'd thought once that policy was a rational thing.

Highlight (Yellow) | Location 9796

A week of hiking, hiking down into the canyon, back up the steep and narrow trails that clung to its walls. Hiking where she could see the age of the planet, as clear as day; where she could see millions of years of history, written in broad stripes of rock and sediment along the sides of the canyon. A place where you could glimpse the scale on which the universe operated. A scale far bigger than the human. A scale that dared humanity to think bigger. To lift its eyes. To reach for more.

Highlight (Yellow) | Location 9882

His mind drifted back to the top secret report he'd read on the events in Shanghai. Varun shook his head in admiration. Lane had stayed true to his principles, all the way through. He'd seen the second order effects, the importance of precedent, of how a history of violence led to more violence, a history of cooperation led to more cooperation. Simple iterated Prisoner's Dilemma. But Lane saw it. Lane put it into action. And he'd engineered cooperation, somehow salvaged it out of the worst possible case of enslavement, torture, and the very edge of the worst possible retribution. Tit for tat on a global scale. Interrupted. Self-sacrifice salvaged out of mutually assured hatred.

Highlight (Yellow) | Location 9936

For now there was work to do. A lot of work. Coercion was still far too easy with Nexus. Emotional manipulation was still far too easy – Breece had shown that with the riots he'd started, the violence he'd incited.

Highlight (Yellow) | Location 9995

The prime obstacle is the hardware. To build something like Nexus one needs a way to interface with millions of neurons at once, and ideally to do so without requiring brain surgery. And I don't actually expect that we'll see something like the Nexus nanites by 2040. That said, various teams are starting to look at ways to move beyond current hardware. At UC Berkeley, Professor Michel Maharbiz and colleagues are working on a project they call "Neural Dust". Their neural dust would be particles that are less than one hundred microns across. Thousands or tens of thousands or more could be sprinkled across the brain, and would then communicate wirelessly via ultrasonic sound waves. Those same ultrasonic vibrations would also provide power to the individual neural dust nodes. Reading their proposal and looking at the diagrams almost feels like reading an artifact from one of my books. Almost. Neural dust still isn't small enough to cross the blood-brain barrier, so it would still involve an incision in the skull. And it's still on the drawing board, not something that has been built. Still, it's fascinating to see science moving in these directions.

Highlight (Yellow) | Location 10005

DARPA, the Defense Applied Research Programs Agency, is also highly interested in this area. In recent talks and calls for grant applications, they've painted a vision of neural interfaces small enough that they could be injected via a syringe rather than via brain surgery; of combining multiple technologies to achieve high-fidelity communication with individual neurons; and most recently of their "cortical modem" vision: an implant the size of two nickels, with a cost of just a few dollars, that would allow the beaming of high-quality video directly into a human brain.

Highlight (Yellow) | Location 10010

Research inside the body and brain moves slowly. The first rule of medicine is, quite appropriately, "do no harm". And that means that experimentation is necessarily conservative. I expect tremendous progress by 2040 – restoration of senses, of mobility, of function destroyed by stroke and brain damage; perhaps even significant augmentation. But I'll be surprised (alas) if it's as easy as drinking a vial of silvery liquid.

Highlight (Yellow) | Location 10015

We all live inside our brains, after all. Our minds are the seat of joy and sorrow, of peace and war, of innovation and stagnation. Unlocking the mysteries of the mind is one of the most exciting and most important endeavors I can imagine.
